MILESTONE 7

SUBMISSION CHECKLIST AND COVER SHEET

DATE: September 23 1997

TO : Dr. Linda Behrens

FROM: TEAM MEMBERS WASHINGTON, ANGELA
RAGHURAMAN, ANANTH

We are submitting the following milestone for your evaluation.

Milestone 7 checklist:

[X] We are resubmitting our System Profile (from Milestone 2).

[ ] We are resubmitting our data flow diagrams (form Milestone 5). We have circled those data flows and processes for which we have completed project repository entries.

[X] We are submitting project repository entries for all of the above data flows.

[X] We are submitting process project repository entries in the form of:

- [X] Structured English
- [ ] Decision Tables
- [ ] Decision Trees
- [X] Other We are listing all the processes
- [ ] We are submitting the following additional documentation:______________________

We have carefully checked our project repository entries for the following:

[X] Each output or process includes a brief description.

[X] We have used descriptive attribute names.

To be completed by the instructor:

I have reviewed Milestone 2. My evaluation is as follows:

[ ]Milestone returned ungraded because of______________________

____

[✓] Milestone acceptable as submitted. Grade of 25
recorded in students' record.

[ ] Milestone acceptable as submitted; however it needs to be corrected and resubmitted to establish a grade.

[ ] Milestone NOT acceptable as submitted. Grade of_______ recorded in students' record.

[ ] Milestone NOT acceptable as submitted because_____________

____

Resubmit by date of_______________ to establish a permanent grade.

Attribute Project Repository Form

Reference Number	Attribute Name	Type	Size	Default value(s)	Edit Rules	Source
01	Student Last Name	Alpha	15	Null		Student
02	Student First Name	Alpha	15	Null		Student
03	Student Middle Initial	Alpha	01	Null		Student
04	Street Address	Alpha-numeric	40	Null		Student
05	City	Alpha	30	Null		Student
06	State	Alpha	02	Null		Student
07	Zip Code	Numeric	08	Null	Format: 99999999	Student
08	Social Security	Alpha-Numeric	11	Null	Format: 999-99-9999	Student
09	Student Pin	Numeric	07	Null	Format: 99-9999	Student
10	Major	Alpha	40	Undecided		Student
11	Credits Earned	Numeric	03	Null	Format: 999	Student
12	Minor	Alpha	40	Null		Student
13	VA or Not	Alpha	02	Null	Format: Y or N	Student
!4	Advisor's Last Name	Alpha	30	Null		Advisor
15	Advisor's First Name	Alpha	30	Dr. Rose		Advisor
16	Advisor's Middle Initial	Alpha	01	Null		Advisor

17	Course Name	Alpha	30	Null		Registration Phone Interface
18	Registration Information	Alpha-Numeric	50	Null		Course
19	Course Number	Numeric	07	Null		Course
20	Instructor"s Name	Alpha	45	Null		Course
21	Student"s Pin	Numeric	07	Null	Format: 99-9999	Registration Phone Interface
22	Student"s Last Name	Alpha	30	Null		Registration Phone Interface
23	Student's First Name	Alpha	30	Null		Registration
24	Department Number	Numeric	09	Null	Format: 99-999-99	Scholastic Organization, Business, and Financial Aid
25	Required Course	Alpha-Numeric	60	Null		Prerequisite
26	Number of Vacant Seats	Numeric	02	Null		Available Course
27	Term	Numeric	06	Null		Presently Registered Course
28	Student's Last Name	Alpha	15	Null		Presently Registered Course
29	Student's First Name	Alpha	15	Null		Presently Registered Course
30	Student's Middle Initial	Alpha	01	Null		Presently Registered Course

31	Minimum of 12 Credits	Alpha	01	Null	Format: Y/N	Requires
32	Room Number	Alpha-Numeric	05	Null	Format: YY-999	Course
33	Student's SSN	Numeric	11	Null		Presently Registered Course
34	Student Telephone Number	Numeric	08	Null	Format: 999-9999	Student
35	Date of Birth	Numeric	08	Null	Format: 99-99-99	Student

List of Project Repository entries for Elementary Processes of Course Addition Subsystem

1.1.3

If Course is available
then
Assign Course by adding Course to student's record (corresponding to PIN) in Student Records.

1.1.2

For each Course number:
1) Using the Course List, search for Course.
2) Find out if it is available by looking up number of vacant seats corresponding to Course Number

1.1.1.1

For each Course number:
1) Using the Course number as key search for course's prerequisite in Prerequisite List.
2) Output the prerequisite to 1.1.1.2

1.1.4

If Course is available (using the result from 1.1.2)
then
tell the student course has been added
else
tell the student course hasn't been added and that it wasn't available.

1.1.1.2

For each prerequisite number and PIN input from 1.1.1.1:
search for the prerequisite in the student's (corresponding to PIN) record, in Student Records.

1.1.1.3

If student has prerequisite
then
proceed to Assign Course
else
tell the student prerequisite isn't available and course hasn't been added.

List of Project Repository entries for Elementary Processes of Course Drop and Information Retrieval Subsystem

1.2.1

If total number of credits after dropping < 12
then
Don't give validation to drop
else
Allow drop by validating attempt.

1.2.2

If validation from 1.2.1 is positive (i.e, can drop course)
then
1) Drop Course by deleting Course in Student record.
2) Relay validation to 1.2.3
else
Don't drop Course.

1.2.3

If Validation is positive
then
Tell Student that Course has been dropped
else
Give reason for not dropping Course.

1.3.2.2 & 1.3.2.1

If User is a University Authority
then
Give him/her access to Information about all Student Records. This includes ability to update information.
else
Give him/her just available Course list

1.3.1

1) Get PIN from User
2) Using PIN as key search Student and University Records to determine User-Type
3) **If** User is Student
then User-Type = Student
else User-Type = University Authority.
4) Input User-Type to 1.3.2

List of Project Repository entries for Elementary Processes of Course Drop and Information Retrieval Subsystem

cont...

1.3.2.2.1

Update Student's Records

1.3.2.2.2

List names and Course numbers of all Courses whose number of vacant seats > 0

1.3.2.2.3

1) For detail report, give names and social security numbers and other registration details of student
2)For exception report give list of all students who are on hold.
3)For summary report, give list according to query.

Project entry for Security Check Subsystem

1.4

For each user:

1) Get PIN
2) Search for PIN in Student and University Authority Records.
2) **If** PIN is found
   then
   allow user to access system
   else
   don't allow user to access system and inform him/her that he/she entered an invalid PIN

Project Repository Entries for Data Flows associated with Elementary Processes

Name	Definition	Trigger
AVAILABILITY	A variable which informs whether a Course is available or not	An attempt by student to add a course.
PRESENCE OF PREREQUISITE	A variable which informs whether a student has taken the required Prerequisite for a Course	An attempt by student to add a course
NUMBER OF VACANT SEATS	A variable which tells how many seats required to be filled up before the course is unavailable	An attempt by Student to add a Course
UPDATE STUDENT RECORD	A change in the information concerning a Student's registration	Updating by University Authority

COURSE NUMBER	A number which uniquely identifies a Course	Attempt to access system, attempt to add Course, attempt to drop Course, attempt to update or retrieve registration information.
PIN	A number which uniquely identifies a User and also provides for secure transactions	Attempt to access system
PREREQUISITE COURSE NUMBER	Same as Course number	Same as Course number
VALIDATION FOR DROP	A variable which tells whether a Course can be dropped.	Attempt to drop Course
REPORT (corresponding to 1.2.3)	Tells the reasons why a Course drop attempt was successful or not	Attempt to drop Course
UNIVERSITY USER TYPE	A variable which allows a University Authority to access Read-and-Write information from Student Records	Attempt to retrieve Information

INFORMATION	Information which includes information about available courses and Student Records	Attempt to retrieve information
READ-AND-WRITE INFORMATION	Information about a student which can be changed or updated	Attempt by University Authority to retrieve or update information
VALIDATION TO ACCESS REGISTRATION SYSTEM	Determines whether a user can access the registration system or not	Attempt to access system

Record entries for the Data Flows of

Elementary Processes

Note:Underlined attributes are
identifiers and italicized
attributes are subrecords.

1. AVAILABILITY =

    COURSE NUMBER +
    [0
    1]

2. PRESENCE OF PREREQUISITE =

    PREREQUISITE COURSE NUMBER +
    [0
    1]

3. NUMBER OF VACANT SEATS =

    COURSE NUMBER +
    [0
    1
    2
    .
    .
    N]

4. UPDATE STUDENT RECORD =

    STUDENT SOCIAL SECURITY NUMBER +
    <COURSE NUMBER,
    CREDITS EARNED,
    MAJOR,
    HOLD,
    TUITION STATUS>

5. VALIDATION FOR DROP =

    STUDENT PIN +
    COURSE NUMBER
    [0
    1]

6. REPORT =

<u>STUDENT PIN</u> +

<u>COURSE NUMBER</u> +

AVAILABILITY OF COURSE +

AVAILABILITY OF PREREQUISITE

7. UNIVERSITY USER TYPE =

<u>PIN</u> +

VALIDATION FOR READ-AND-WRITE INFORMATION

8. STUDENT USER TYPE =

<u>PIN</u> +

VALIDATION TO ACCESS READ-ONLY INFORMATION +

LIST OF COURSE NUMBERS OF AVAILABLE COURSES

9. INFORMATION =

<u>PIN</u> +

READ-ONLY INFORMATION +

READ-AND-WRITE INFORMATION

10. VALIDATION TO ACCESS REGISTRATION SYSTEM =

<u>PIN</u> +

[0

1]

Project Repository Entries for Outputs

TYPE: STUDENT REPORT

EXPLODES TO : REPORT ON COURSE DROP (page 2)

REPORT ON COURSE ADD (page 3)

INFORMATION RETRIEVAL (page 4)

DESCRIPTION : THIS REPORT IS GENERATED WHEN THE STUDENT ATTEMPTS TO ADD OR DROP COURSES OR WHEN HE/SHE GETS INFORMATION ON AVAILABLE COURSES.

MEDIUM: PHONE INTERFACE

OUTPUT CHARACTERESTICS: IT IS A DATA FLOW FROM THE COURSE ADDITION, COURSE DROP AND INFORAMTION SYBSYSTEMS. AS TO INFORMATION RETRIEVAL, THE STUDENT CAN ACCESS READ-ONLY INFORMATION CONCERNING AVAILABLE COURSES AND HIS/HER OWN RECORDS. THE REPORT IS VOCAL, COMMUNICATED BY A RECORDED VOICE.

OUTPUT COMPOSITION: COURSE NUMBERS

REASONS FOR SUCCESS IN ATTEMPT OR LACK OF IT

TYPE: REPORT ON COURSE DROP

DESCRIPTION: THIS REPORT IS GENERATED WHEN THE STUDENT ATTEMPTS TO DROP COURSES. IT TELLS THE STUDENT WHETHER A COURSE WAS DROPPED OR NOT AND THE REASONS FOR NOT DROPPING A COURSE.

MEDIUM: PHONE INTERFACE

OUTPUT CHARACTERESTICS: IT IS A DATA FLOW FROM COURSE DROP SUBSYSTEM TO STUDENT. IT IS IN THE FORM OF A RECORDED VOICE. IT IS GENERATED IN THE COMPUTER CENTER OF THE UNIVERSITY. IT OCCURS WITH EVERY INSTANCE WHERE A STUDENT TRIES TO DROP A COURSE.

OUTPUT COMPOSITION: COURSE NUMBERS OF COURSES DROPPED
REASONS FOR NOT DROPPING A COURSE

TYPE: REPORT ON COURSE ADD

DESCRIPTION: THIS REPORT IS GENERATED WHEN A STUDENT ATTEMPTS TO ADD COURSES. IT TELLS THE STUDENT WHETHER A COURSE WAS ADDED OR NOT AND THE REASONS FOR NOT ADDING A COURSE.

MEDIUM: PHONE INTERFACE

OUTPUT CHARACTERESTICS: IT IS A DATA FLOW FROM COURSE ADDTION SUBSYSTEM TO STUDENT. IT IS IN THE FORM OF A RECORDED VOICE. IT IS GENERATED IN THE COMPUTER CENTER OF THE UNIVERSITY. IT OCCURS WITH EVERY INSTANCE IN WHICH A STUDENT ATTEMPTS TO ADD A COURSE.

OUTPUT COMPOSITION: COURSE NUMBERS

REASONS FOR ADDING/NOT ADDING A COURSE

TYPE:	INFORMATION RETRIEVAL
EXPLODES TO:	READ-AND-WRITE INFORMATION REPORT (page 5)
	READ-ONLY INFORMATION REPORT (page 6)
MEDIUM:	PHONE INTERFACE AND COMPUTER TERMINALS

OUTPUT

CHARACTERESTICS: IT IS A DATA FLOW FROM INFORMATION RETRIEVAL SUBSYSTEM TO THE STUDENT. IT IS GENERATED IN THE FORM OF A RECORDED VOICE, IN CASE A STUDENT IS USING A PHONE AND COMPUTER SCREEN IN CASE A UNIVERSITY AUTHORITY IS USING A TERMINAL.

OUTPUT

COMPOSITION: COURSE NUMBERS OF AVAILABLE COURSE,
COURSE NUMBERS OF UPDATED COURSES,
SOCIAL SECURITY NUMBERS OF STUDENTS WHOSE RECORD HAS BEEN UPDATED.

TYPE:	READ-AND-WRITE INFORMATION
EXPLODES TO:	DETAIL REPORT (page 7)
	SUMMARY REPORT (page 8)
	EXCEPTION REPORT (page 9)
MEDIUM:	COMPUTER TERMINAL
OUTPUT CHARACTERESTICS:	IT IS A DATA FLOW FROM INFORMATION RETRIEVAL SUBSYSTEM TO STUDENT. IT IS IN THE FORM OF COMPUTER SCREENS. IT IS GENERATED IN THE COMPUTER CENTER. IT OCCURS WITH EVERY INSTANCE IN WHICH A UNIVERSITY AUTHORITY ATTEMPTS TO RETRIEVE REPORTS OR STUDENT RECORDS.
OUTPUT COMPOSITION:	STUDENT RECORDS

TYPE: READ-ONLY-INFORMATION REPORT

DESCRIPTION: THIS REPORT IS GENERATED WHEN A STUDENT ACCESSES THE SYSTEM TO GET INFORMATION. IT GIVES HIM/HER THE LIST OF AVAILABLE COURSES, INFORMATION ON HIS/HER OWN RECORDS.

MEDIUM: PHONE INTERFACE

OUTPUT

CHARACTERESTICS: IT IS A DATA FLOW FROM INFORMATION RETRIEVAL SUBSYSTEM TO THE STUDENT. IT IS IN THE FORM OF A RECORDED VOICE. IT IS GENERATED IN THE COMPUTER CENTER OF THE UNIVERSITY. IT OCCURS WITH EVERY INSTANCE WHERE A STUDENT ACCESS THE SYSTEM TO RETREIVE INFORMATION.

OUTPUT

COMPOSITION: COURSE NUMBERS OF AVAILABLE COURSES, INFORMATION ON HIS/HER OWN RECORD.

TYPE:	DETAIL REPORT
DESCRIPTION:	THIS REPORT GIVES THE UNIVERSITY AUTHORITY, REGISTRATION INFORMATION ON ALL STUDENTS AND COURSES.
MEDIUM:	COMPUTER TERMINAL
OUTPUT CHARACTERESTIC:	IT IS A DATA FLOW FROM INFORMATION RETRIEVAL SUBSYSTEM TO STUDENT. IT IS IN THE FORM OF COMPUTER SCREENS. IT IS GENERATED IN THE COMPUTER CEWNTER.
OUTPUT COMPOSITION:	STUDENT RECORDS COURSE LIST, PREREQUISITE LIST

TYPE:	SUMMARY REPORT
DESCRIPTION:	THIS REPORT SUMMARIZES THE REGISTRATION INFORMATION OF EACH STUDENT AND GIVES ONLY HIS/HER MAJOR, CREDITS EARNED, GPA AND TUITION STATUS (WHETHER THE STUDENT PAID HIS/HER TUITION OR NOT).
MEDIUM:	COMPUTER TERMINAL
OUTPUT CHARACTERESTICS:	IT IS A DATA FLOW FROM INFORMATION RETRIEVAL SUBSYSTEM TO UNIVERSITY AUTHORITY. IT IS IN THE FORM OF COMPUTER SCREENS.
OUTPUT COMPOSITION:	STUDENT RECORDS

TYPE:	EXCEPTION REPORT
DESCRIPTION:	DEPENDING UPON THE CRITERIA, THIS GIVES ALL THE STUDENT RECORDS WHICH SATISFY THE CRITERIA.
MEDIUM:	COMPUTER TERMINAL
OUTPUT CHARACTERESTICS:	IT IS A DATA FLOW FROM INFORMATION RETRIEVAL SUBSYSTEM TO UNIVERSITY AUTHORITY. IT IS GENERATED IN THE COMPUTER CENTER.
OUTPUT COMPOSITION:	STUDENT RECORDS

Project Repository Entries for Inputs

TYPE:	PIN
DESCRIPTION:	IT IS A NUMBER GIVEN TO A STUDENT WHICH UNIQUELY INDENTIFIES A USER AS WELL AS DECIDES WHAT TYPE OF INFORMATION HE/SHE ACCESS
MEDIUM:	PHONE INTERFACE/COMPUTER TERMINAL
INPUT CHARACTERESTICS:	IT IS A DATA FLOW FROM THE STUDENT TO THE REGISTRATION SYSTEM.
INPUT COMPOSITION:	FOUR-DIGIT NUMBER

TYPE:	COURSE NUMBER
DESCRIPTION:	THIS NUMBER UNIQUELY IDENTIFIES A COURSE
MEDIUM:	PHONE INTERFACE / COMPUTER TERMINAL
INPUT CHARACTERESTICS:	IT IS A DATA FLOW FROM STUDENT/UNIVERSITY AUTHORITY TO THE SYSTEM. THE USER ENTERS IT BY PUNCHING IN THE NUMBER THROUGH THE KEY PAD OF A TOUCH-TONE PHONE.
INPUT COMPOSITION:	8 DIGIT NUMBER

MILESTONE 8

SUBMISSION CHECKLIST AND COVER SHEET

DATE: November 5, 1997

TO : Dr. Linda Behrens

FROM: TEAM MEMBERS WASHINGTON, ANGELA
RAGHURAMAN, ANANTH

We are submitting the following milestone for your evaluation.

Milestone 8 checklist:

[ ] We are resubmitting our System Profile for your reference.

[ ] We are resubmitting our original entity relationship diagram for your reference.

[X] We are submitting a list of data attributes, categorized by data entity, that has been normalized to 3rd normal form.

[X] As a result of performing normalization, we are submitting a new, revised entity relationship, which reflect changes caused by normalization (i.e. new entities).

[X] We are submitting an event analysis, we are submitting new, revised data flow diagrams, which reflect changes by the event analysis (i.e. additional data flows from data stores).

[X] We are submitting the following additional documentation:__

____

We have carefully checked the following:

[X] Each entity truly describes the entity into which we have placed it. The attribute's value is dependent upon the key to that entity, all parts of that key (if it is a combination key), and nothing else but the key.

[X] For each relationship on our entity relationship diagram, we have identified which entity entity is the parent and which entity is the child.

[X] For each event in our event analysis, we have specified the entity affected, the event name and description, the action that causes (to the entity), and any business or data model rules or conditions that apply to that event.

To be completed by the instructor:

I have reviewed Milestone 8. My evaluation follows:

[ ] Milestone returned ungraded because of______________________

____

[ ] Milestone acceptable as submitted. Grade of 85___ recorded in students' record.

[ ] Milestone acceptable as submitted; however, it needs to be corrected and resubmitted to establish a permanent grade.

[ ] Milestone **NOT** acceptable as submitted. Grade of____________ recorded in student's record.

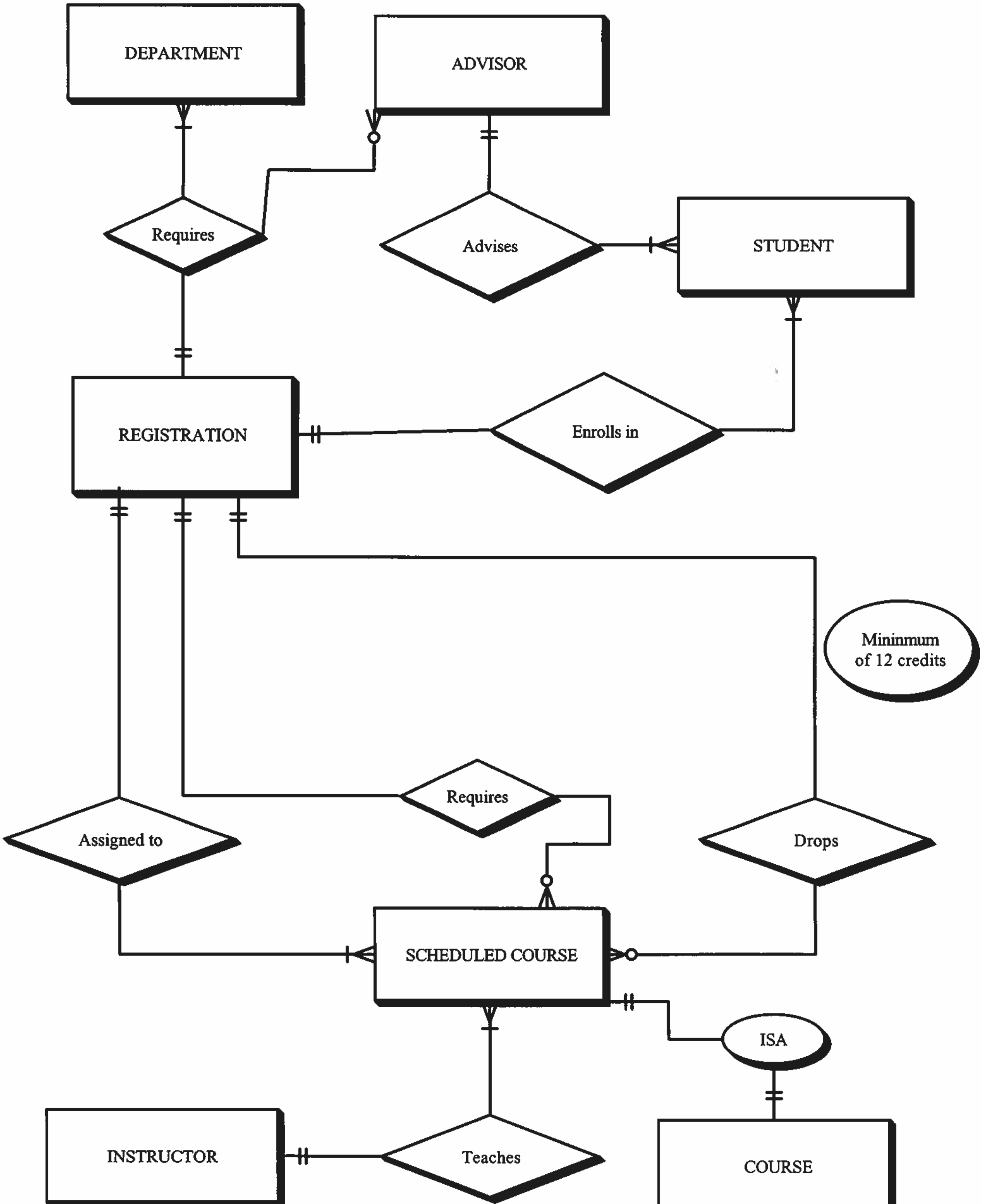
[√] Milestone **NOT** acceptable as submitted because______________

____

Resubmit by date of________________________ to establish a permanent grade.

11-10 -5 cover
-10 late
~~-10 lates~~

NORMALIZED ERD (All attributes are listed on the next page)



EVENT RESPONSE LIST

EVENT	TRIGGER (INPUTS)	RESPONSES (OUTPUTS)
COURSE ADDITION	Student PIN, Course numbers	Report, Course addition
COURSE DROP	Student PIN, Course numbers	Report, Course drop
UPDATE STUDENT RECORDS	University authority's PIN	Read-and-Write information
GIVE READ-ONLY INFORMATION	Student PIN	Read-Only information

Attributes of Entities

ADVISOR

- Advisor Last Name
- Advisor Social Security Number
- Advisor First Name

STUDENT

- Student PIN
- Student Social Security Number
- Student Last Name
- Student First Name
- Student Middle Initial
- Student Type (whether new/readmitted/continuing)
- Enrollment Term (whether interterm/spring/summer I/ Summer II/fall/spring)
- Race (whether white-non Hispanic/black-non Hispanic/Hispanic/Asian,Pacific Islander/non-resident alien/Amer. Indian/Alaska native)
- Major
- Minor
- Sex
- Date Of Birth
- Temporary Address
- Permanent address
- Phone Number
- Credits Earned
- VA Status
- Day Student (0 if not 1 if yes)
- Degree Status (whether Non-Degree/MBA/Nursing/MAT,MED)
- CSU Athletic Status (Baseball/Basketball/Cross Country/Football/Golf/Soccer/Softball/ Tennis/Track/field/Volleyball)

SCHEDULED COURSE

- Section Number
- Course Number
- Room Number
- Building Number
- Start Time
- End Time
- Days Scheduled
- Instructor Last Name
- Instructor First Name

COURSE

- Course Number
- Course Title
- Number Of Credits
- Department Number

DEPARTMENT

- Department Number
- Department Name

REGISTRATION

- Course Number
- Section Number
- Student Social Security Number

INSTRUCTOR

- Instructor Social Security Number
- Instructor Last Name
- Instructor First Name
- Instructor Degree
- Office Address
- Phone Number
- Department Number (with which the professor is associated)

Entity Processes CRUD

Entity,attribute / **Process**	Update Student Records	Course Addition	Course Drop	Give Read-Only information (to student)
ADVISOR	X	X	X	R
Advisor First name	X	X	X	X
Advisor Last name	X	X	X	X
Advisor Social security number	X	X	X	X
STUDENT	U	U	U	R
Student Last name	U	X	X	X
Student First name	U	X	X	X
Student PIN	C	X	X	X
Credits earned	U	U	U	X
Temporary address	U	X	X	R
Student type	U	X	X	R
Major	U	X	X	X
Phone number	U	X	X	X
Date of birth	C	X	X	X
Minor	U	X	X	X
Student Social security number	C	X	X	X
Race	C	X	X	X
Enrollment term	U	X	X	X
VA status	C	X	X	X
Day student	U	X	X	X
Degree status	U	X	X	X
CSU Athletic status	U	X	X	X
Student Midle initial	C	X	X	X
Sex	C	X	X	X
Permanent Address	U	X	X	R

Entity, attributes \ Process	Update Student records	Course Addition	Course Drop	Read-Only information (to student)
SCHEDULED COURSE	X	X	X	X
Course number	X	X	X	R
Section number	X	X	X	R
Room Number	X	X	X	R
Building Number	X	X	X	R
Start Time	X	X	X	R
End Time	X	X	X	R
Days Scheduled	X	X	X	R
Instructor Last Name	X	X	X	R
DEPARTMENT	X	X	X	R
Department Name	X	X	X	R
Department Number	X	X	X	R
REGISTRATION	U	U	U	X
Course number	U	U	U	X
Student Social security number	X	X	X	R
Section Number	U	U	U	X
COURSE	X	X	X	X
Course Number	X	X	X	R
Course Title	X	X	X	R
Number of Credits	X	X	X	R
Department Number	X	X	X	R

MILESTONE 9

PROCESS ANALYSIS AND DESIGN

MILESTONE 9

SUBMISSION CHECKLIST AND COVER SHEET

DATE: November 6,1997

TO : Dr. Linda Behrens

FROM: TEAM MEMBERS WASHINGTON, ANGELA
RAGHURAMAN, ANANTH

We are submitting the following milestone for your evaluation

Milestone 9 checklist:

[ ] We are resubmitting our System Profile for your reference.

[ ] We are resubmitting our data flow diagrams for your reference. We have drawn boundaries on the diagrams to clearly demonstrate the automated and manual portions of the system.

[X] We are submitting: [X] System flowcharts [X] implementation data flow diagrams for documenting the design units in our system.

[X] We are submitting a legend or reader's guide for our diagrams.

[X] Our methods and procedures documentation is supplemented with brief written narratives (one page or less per diagram).

[ ] We are submitting the following additional documentation:___

___

We have carefully checked our diagrams for the following:

[X] Our diagrams generally read from top-to-bottom, left-to-right.

[X] Our inputs, outputs, and files carry the same names on our diagrams that they did in Milestone 5.

[X] We have connected related pages to show continuation of flow (using off-page connectors).

[X] We added file and data base maintenance programs to our diagrams.

[X] All programs have identifiable input symbols and output symbols.

[X] Our procedures show how input errors are corrected by either data entry staff or system users.

[X] Our batch processing procedures include appropriate batch control checks and controls.

[X] Our procedures show how reports get collected and distributed to users and management.

Note: Some of the above are more appropriate to the system flowchart option of this milestone and may be waived if your are using implementation data flow diagrams.

To be completed by the instructor:

I have reviewed Milestone 9. My evaluation follows:

[ ] Milestone returned ungraded because of______________________

____

[ ] Milestone acceptable as submitted. Grade of 40________ recorded in students' record.

[ ] Milestone acceptable as submitted; however, it needs to be corrected and resubmitted to establish a permanent grade.

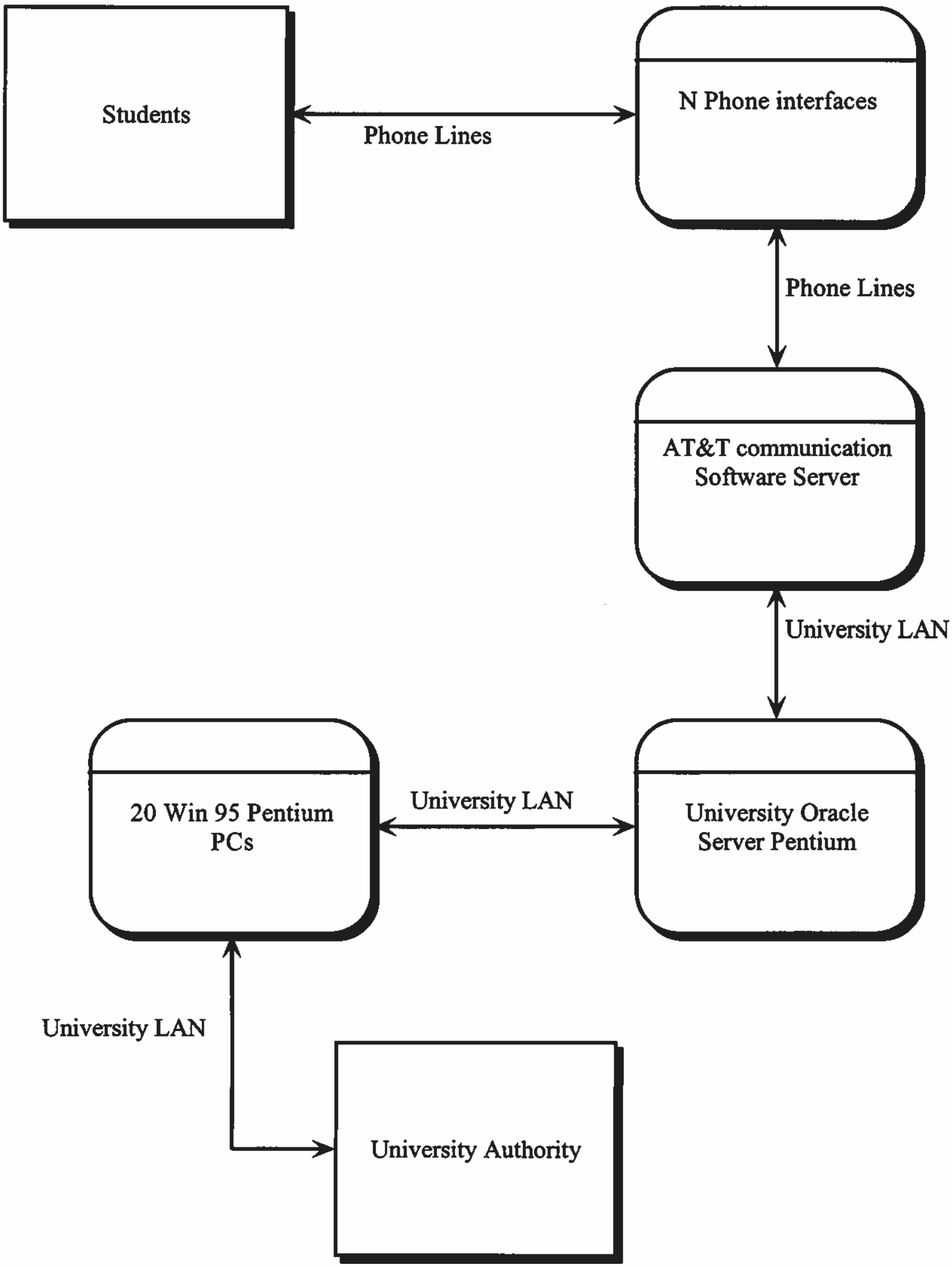
[ ] Milestone **NOT** acceptable as submitted. Grade of __________ recorded in students' record.

[ ] Milestone **NOT** acceptable as submitted because ____________

____

Resubmit by date of__________________ to establish a permanent grade.

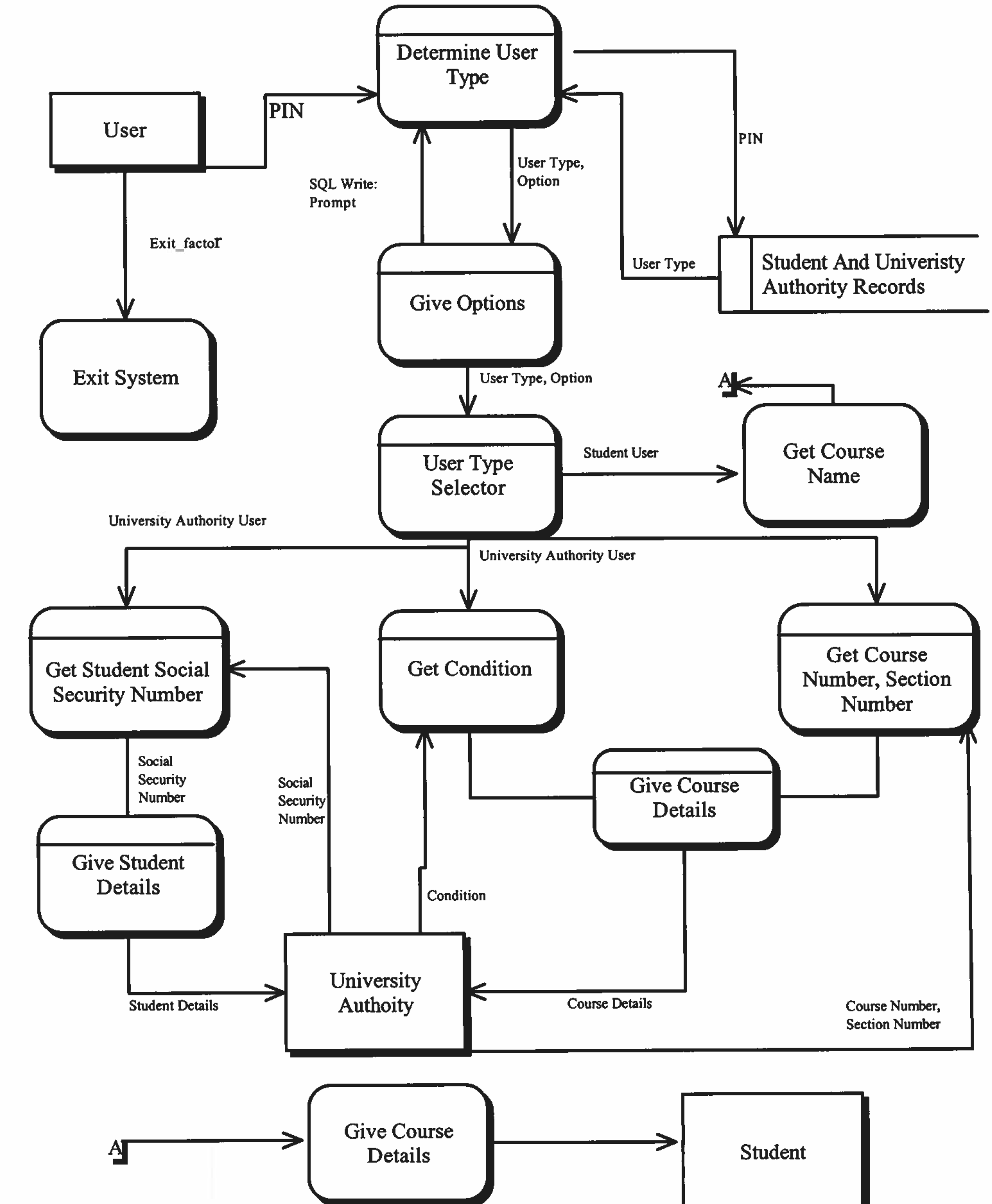
NETWORK TOPOLOGY DFD



Brief narrative of Network Topology DFD

In the diagram, there are two entities: Students and University Authorities. The Students may be located anyplace, but University Authorities are located on-campus. The Students communicate with the Central Oracle Server, which accepts calls to drop/add courses and retrieve information, via phone lines. The AT&T Communication Software converts the phone signals to computer signals while accepting multiple calls at the same time. For the use of University Authorities, there are 20 PCS which are connected to the Central Oracle Server that they can use to retrieve information.

Information Retrieval *Physical* DFD

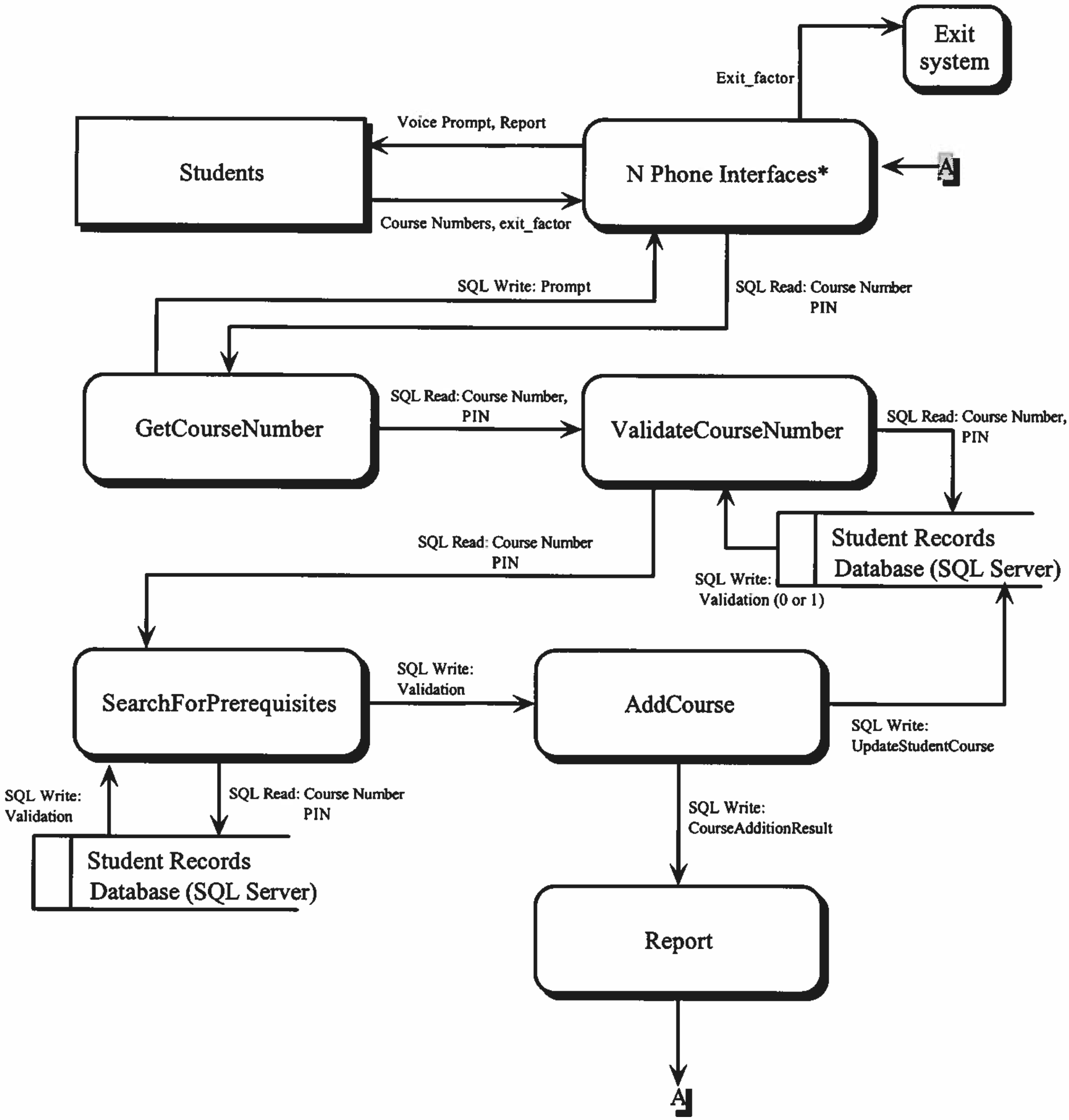


Note: The Process "Give Course Details" utilizes the Course List Database on the University Server Also, all data flows are in SQL.

Brief Narrative of Information Retrieval

In the diagram, the User enters the User's Pin and the system recognizes the User as a Student or an University Authority. After the system recognizes the User, the Voice Prompts the user's options for that type of User. The Student gets the name of a Course and all the information about that Course. If the User is an University Authority, the official can enter the Student's Social Security Number and retrieve all the necessary information about that Student, or the official can enter the Course Number and Section to retrieve all the necessary information about that Course.

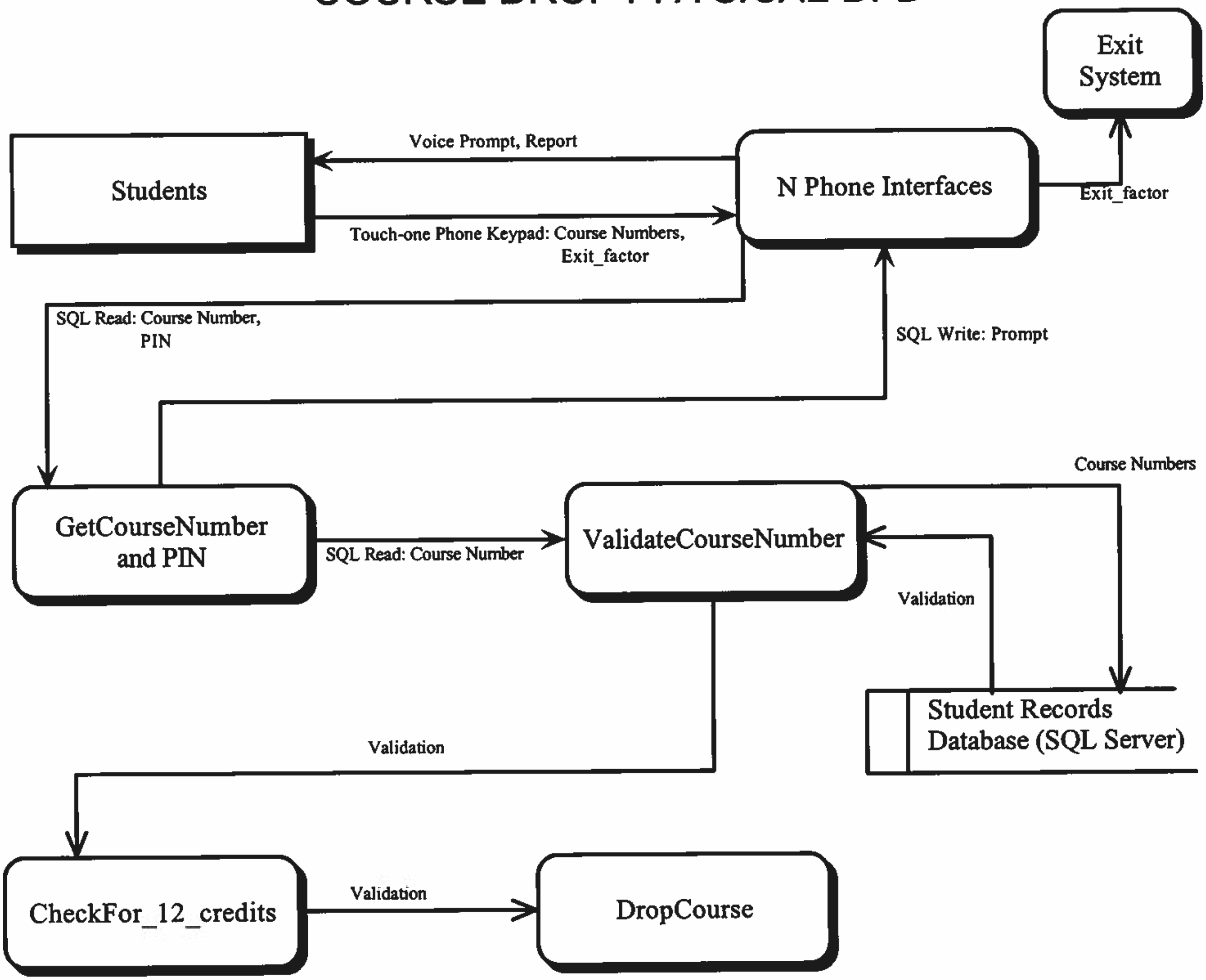
COURSE ADDITION *PHYSICAL* DFD



Brief Narrative of Course Addition

The diagram shows Students (denoted by the square box) adding a course. The data flow associated with Students are, Voice Prompt, Report and Course Numbers. The Voice Prompt guides Students through the process of adding a course. So it is shown as an output. Report is generated when the system has either added the course or not. It tells the student whether the course was added. It also gives reasons for not adding a course. The Students input Course Numbers of the courses they want to register for. Next in the DFD is the Phone interface; it consists of touch-tone which the Students use, and the AT&T communication Server which connects the Oracle Server and phone lines. It acts as interface to the University LAN and phones. The phone interfaces pass on the Students' PIN and the Course Numbers entered, to the Oracle Server. The server has a set of programs, which add courses. The first of these is GetCourseNumber. This is asks the student for the Course Numbers of the courses to be added. After receiving the Course Numbers, it passes them on to ValidateCourseNumber, which checks to see if the student entered the correct Course Number, against Student Records. After validation, the program SearchForPrerequisites checks whether the student has the necessary prerequisite courses. If so, the program AddCourse adds the courses to the Student's list of courses.

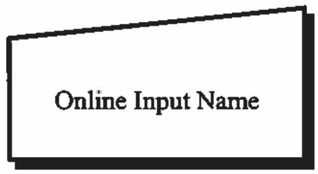
COURSE DROP *PHYSICAL* DFD



Brief Narrative of Course Drop Subsystem

In the diagram, the Student uses a phone to key in the Course Number of the courses they want to drop. Student is guided by the Voice Prompt. The data is then transmitted over phone lines to the logic based in the University Oracle Server. This set of logic consists of the following programs: GetCourseNumber, ValidateCourseNumber, Check_For_12_Credits, and DropCourse. GetCourseNumber prompts the Student for Course Number and Pin, and it passes on the information to ValidateCourseNumber, which checks if the Course Number input is valid, by checking it against the database of Courses. If valid, the program Check_For_12_Credits calculates to see whether the Student will have 12 credits after student drops the course. If so, the program drops the course from the Student's List of courses. If any of these programs don't find a valid reason to proceed, they abort the Course Drop and inform the Student about reasons for doing so. An automatic backup is created when the updates take place.

Legend For System Flowcharts



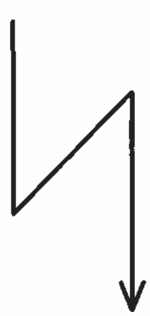
An Online Input

A Program

An Online Output

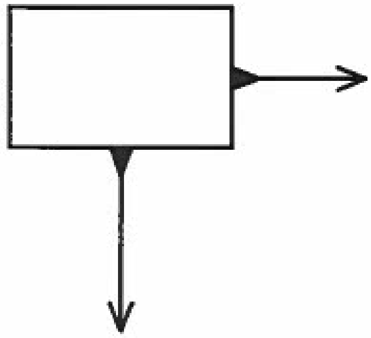
A Disk Drive

Sequence of Activities



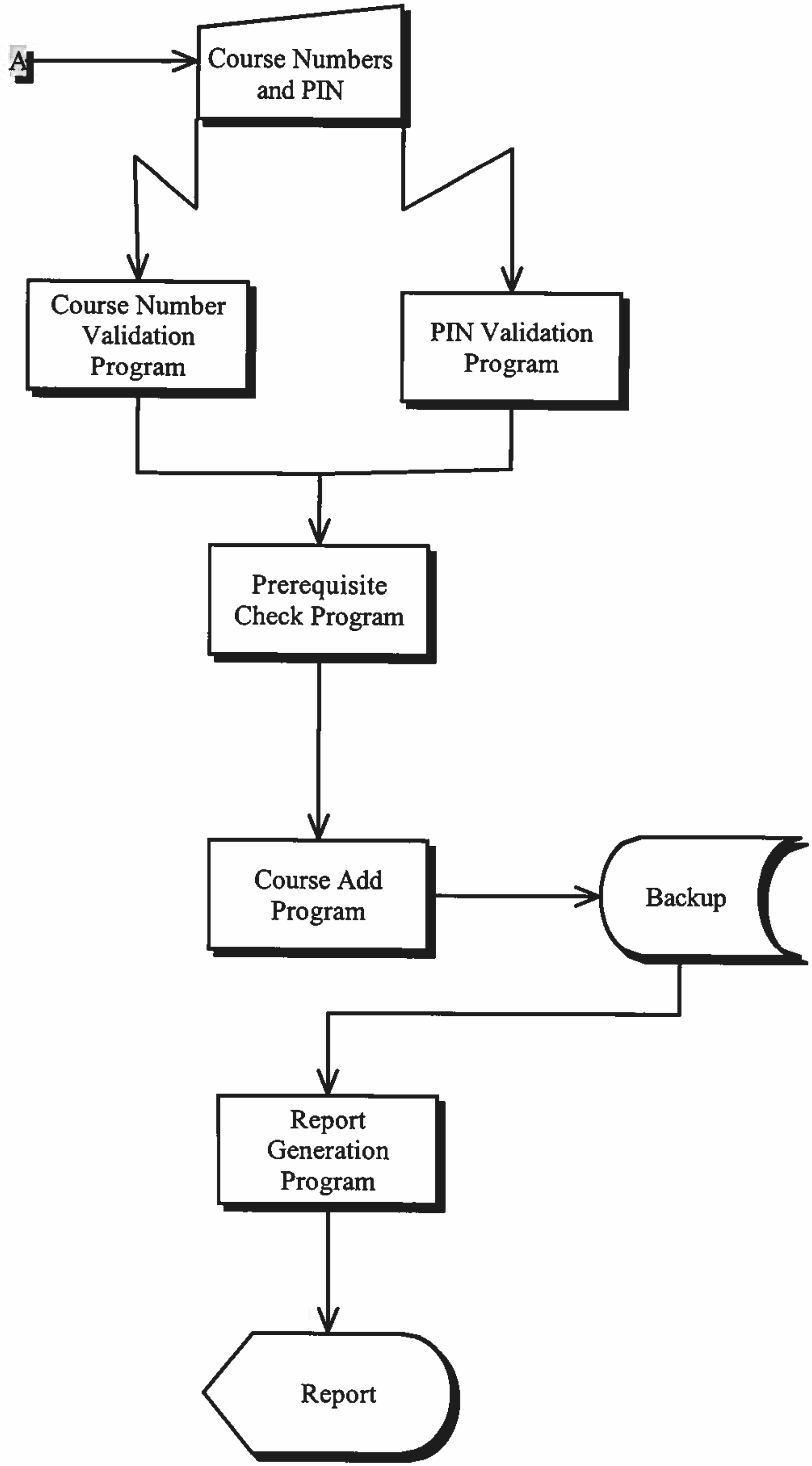
Communication/Data Upload/Download

Legend cont...

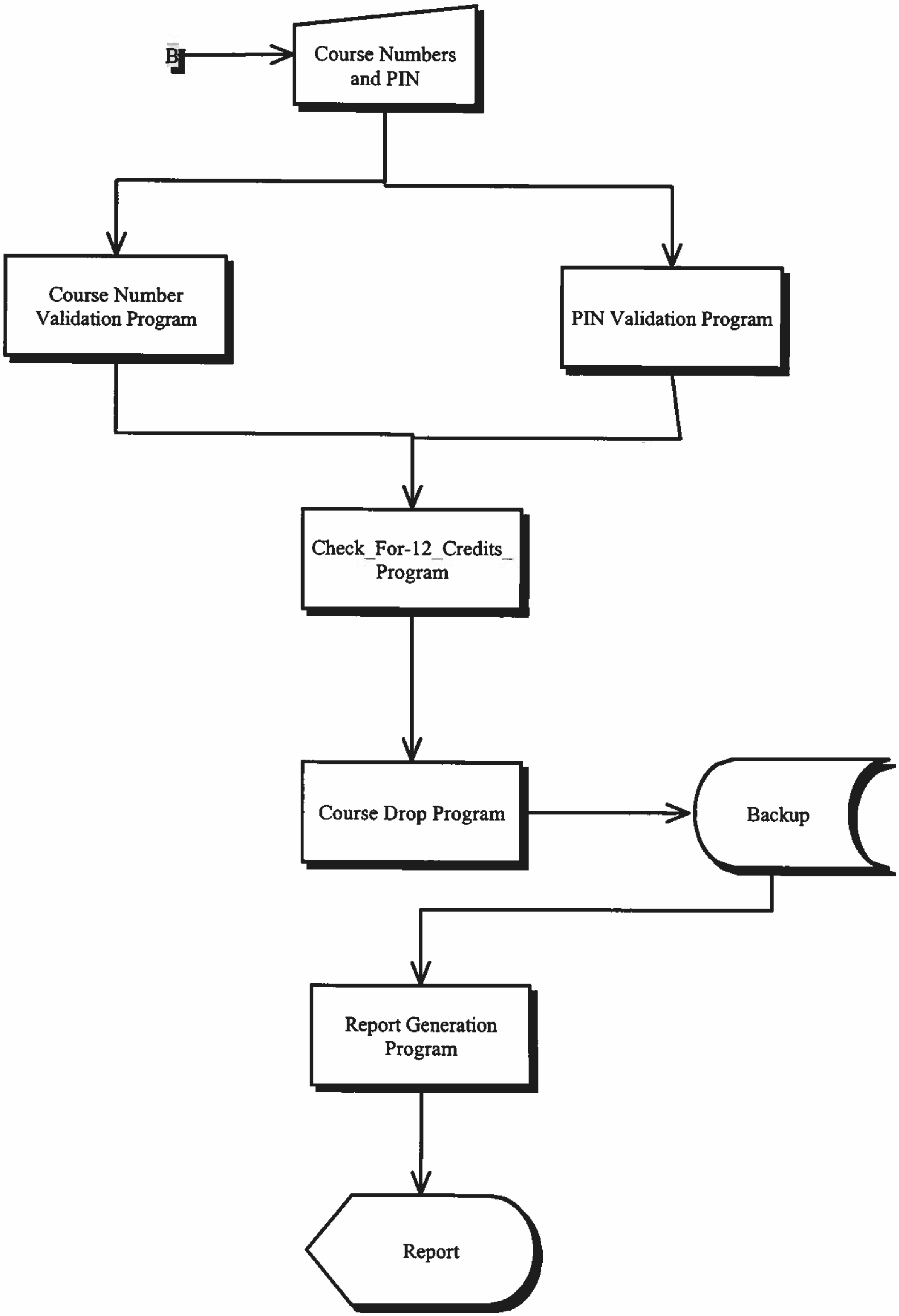


Denotes Either One Or Other Sequence

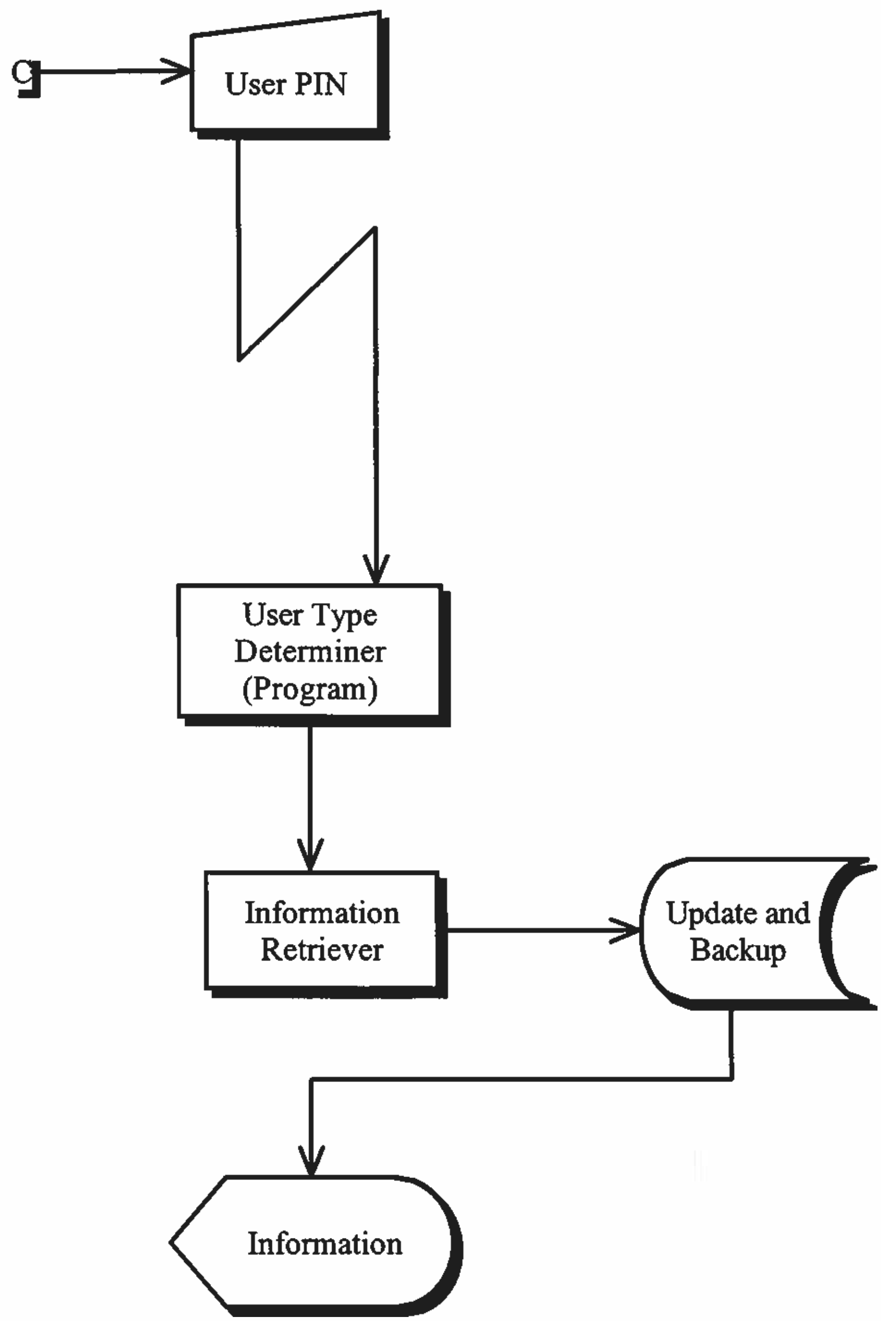
Course Addition System Flowchart



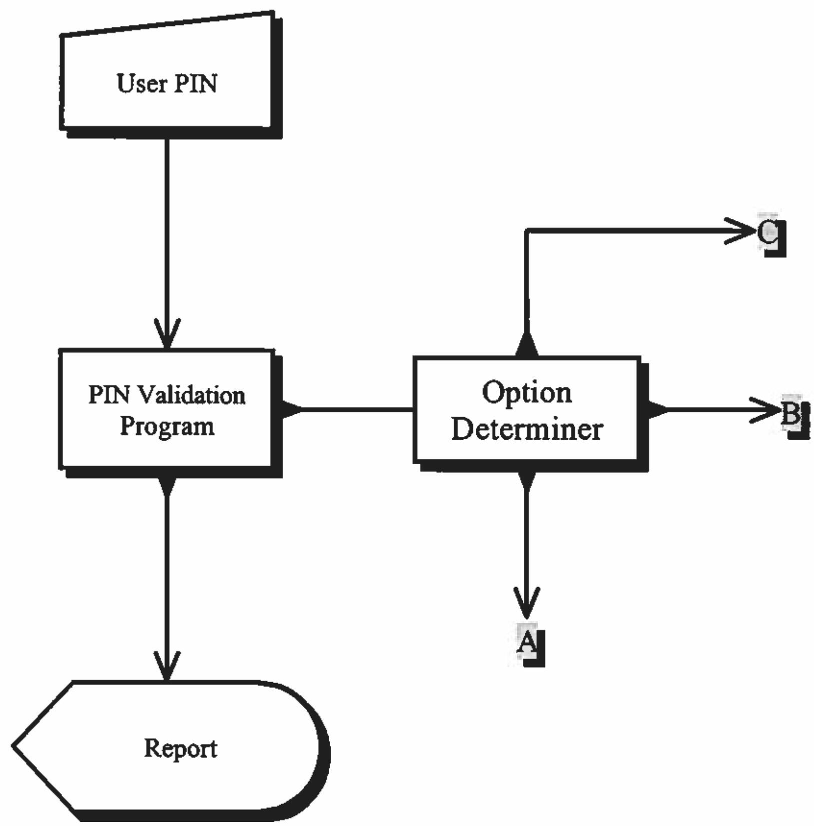
Course Drop System Flowchart



Information Retrieval



Security Check System Flowchart



MILESTONE 10

SUBMISSION CHECKLIST AND COVER SHEET

DATE: November 17, 1997

TO : Dr. Linda Behrens

FROM: TEAM MEMBERS WASHINGTON, ANGELA
RAGHURAMAN, ANANTH

We are submitting the following milestone for your evaluation

Milestone 10 checklist:

[ ] We are resubmitting our System Profile for your reference.

[ ] We are resubmitting our data flow diagrams that contain the files that we designed. We circled those files appearing on the data flow diagram to indicate which files were designed.

[ ] We are submitting expanded project repository entries for each files we designed.

[ ] We are submitting Record Layout Charts for each file.

[ ] We are submitting sample program listing or documentation describing computer files that are representative of our system's files.

[ ] We are submitting test data for our files.

[ ] Our files design specifications are accompanied by a brief explanation of the rationale behind our file design decisions.

[X] We are submitting a data base schema. The data base management system for this schema is: Oracle

[X] We are submitting the data definition language for the above schema.

[ ] We are submitting the following additional documentation:___

____

We have carefully checked our diagrams for the following:

[X] The name of each of our files coincides with the name or title we used on the data flow diagram. We also included aliases for those files to which users commonly refer by different names.

[X] We established codes for appropriate attributes to optimize storage requirements.

[X] We specified the optimal storage format for fields in the logical records and determined size requirement for the file according to the constraints imposed by the computer system.

[X] We determined the required file access methods of each program using the file to select the best file organization.

[X] We minimized data redundancy in designing our files.

[X] We installed internal controls for our files including: access, retention, backup, and recovery controls.

[X] For each attribute that appears in the logical record, we specified the size, the storage format, storage size in bytes, and from where the value of the attribute comes.

[ ] Special comments to the instructor: ______________________

____

To be completed by the instructor:

I have reviewed Milestone 10. My evaluation follows:

[ ] Milestone returned ungraded because of ______________________

____

[✓] Milestone acceptable as submitted. Grade of 40 ________ recorded in students' record.

[ ] Milestone acceptable as submitted; however, it needs to be corrected and resubmitted to establish a permanent grade.

[ ] Milestone **NOT** acceptable as submitted. Grade of __________ recorded in students' record.

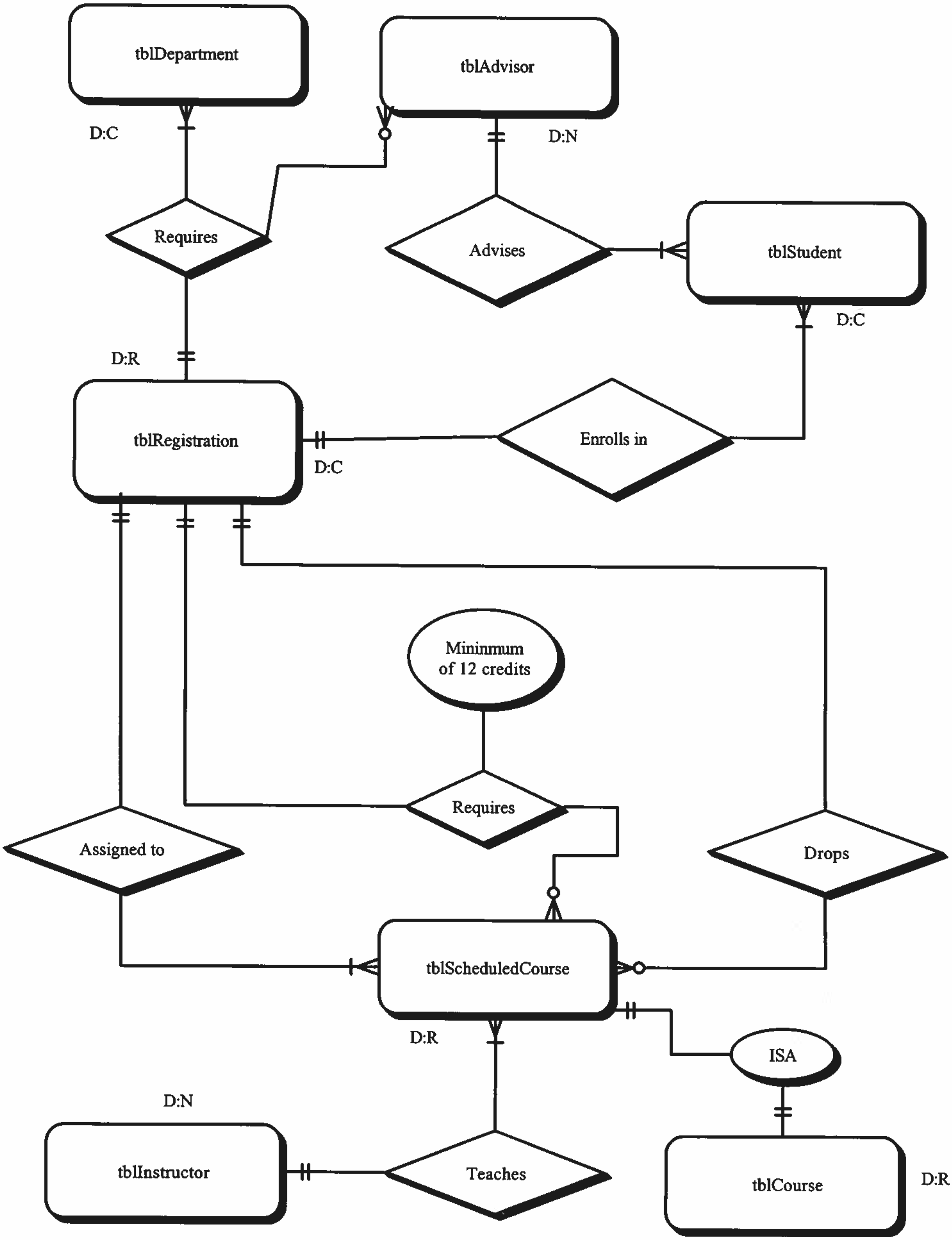
[ ] Milestone **NOT** acceptable as submitted because ____________

____

Resubmit by date of ________________ to establish a permanent grade.

Physical Database Schema

Note: All attributes are listed on the next page due to paucity of space



Attributes of Entities

(Physical Database Schema cont...)

tblDepartment		
colDepartmentNumber	varchar(2)	[PK]
colDepartmentName	varchar(10)	

tblAdvisor		
colAdvisorLastName	varchar(30)	[PK1]
colAdvisorSocial	varchar(11)	[PK2]
colAdvisorFirstName	varchar(15)	

Attributes of Entities cont…

tblStudent		
Physical Display		
colStudentPIN	varchar(4)	[PK1]
colStudentSocial	varchar(11)	[PK2]
colStudentLastName	varchar(30)	
colStudentFirstName	varchar(15)	
colStudentType	varchar(4)	
colTerm	varchar(5)	
colRace	varchar(10)	
colMajor	varchar(9)	
colMinor	varchar(9)	
colSex	char(1)	
colDOB	varchar(8)	
colTempAdd	varchar(40)	
colPermAdd	varchar(40)	
colPhNumber	varchar(10)	
colCrEarned	int	
colVAStatus	int	
colDayStudent	int	
colDegreeStatus	int	
colCSUAthlete	varchar(15)	

Attributes of Entities cont…

tblCourse

Physical Display

colCourseNumber	varchar(2)	[PK]
colCourseTitle	varchar(10)	
colNumberOfCr	int	
colDeptNumber	varchar(2)	[FK]

tblRegistration

Physical Display

colCourseNumber	varchar(2)	[PK] [FK]
colSectionNumber	varchar(2)	[PK] [FK1]
colStudentSocial	varchar(11)	[PK] [FK2]

Note: Please see next page for relationships.

The table Registration is related to:

1. table Student by key StudentSocial
2. table Course by key CourseNumber

Attributes of Entities cont...

tblInstructor Physical Display		
colInstSocial	varchar(11)	[PK]
colInstLastName	varchar(30)	[PK1]
colInstFirstName	varchar(15)	
colInstDegree	varchar(30)	
colIstOffice	varchar(40)	
colInstPhNumber	varchar(10)	
colDeptNumber	varchar(2)	[FK]

Attributes of Entities cont…

tblScheduledCourse Physical Display		
colSectionNumber	varchar(2)	[PK1]
colCourseNumber	varchar(2)	[PK2] [FK]
colRoomNumber	varchar(2)	
colBuildingNumber	varchar(2)	
colStartTime	varchar(5)	
colEndTime	varchar(5)	
colDaysSch	varchar(14)	
colInstLastName	varchar(30)	[FK1]
colInstFirstName	varchar(15)	

The above table is related to the table Instructor by the foreign key, InstLastName and to the table Course by foreign key CourseNumber.

3. table Scheduled Course by key SectionNumber

The table Course is related to table Department by foreign key DepartmentNumber